Synthetic Ferrimagnetic Free Layer for Quasi-differential HDD Reader

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The maturity of recording technology in hard disk drives (HDDs) has stimulated the development of readout technologies compatible with high recording density. Differential readers are expected to be suitable for improving resolution in the downtrack direction, but their adoption in products has been hindered by not only their structural complexity but also drawbacks in long-pattern performance. To address these issues, we consider a laminated free layer (FL) structure in which two unbalanced FLs are antiferromagnetically coupled, forming a synthetic ferrimagnetic (SyF)-FL. A differential signal can be introduced simply with the additional deposition of a spacer and a second FL on top of a conventional single-FL structure. This intentional asymmetry enables finite signal output even away from transitions, resulting in a quasi-differential waveform that maintains long-pattern performance. Although an increase in magnetic thermal noise appears to be another challenge, the SyF-FL reader can provide a practical route to implementing differential readers in HDDs with high recording density.

Index Terms— **Hard disk drive (HDD), magnetic recording, magnetic sensor, spintronics, tunnel magnetoresistance (TMR)**

I. Introduction

Development of energy-assisted magnetic recording technologies, in which the recording process is assisted by heat, microwave, or other effects, has successfully increased the recording density of hard disk drives (HDDs) [1, 2, 3]. As a result, readout technology now faces the challenge of reading increasingly small recording patterns, and readers with higher spatial resolution are required.

The foundation of HDD readers is the spin-valve (SV) structure, in which the relative angle between a free layer (FL) and a pinned layer (PL) is converted into an electrical signal via the tunnel magnetoresistance (TMR) effect [4]. Significant effort has been devoted to scaling down reader dimensions to achieve higher resolution. In particular, reducing the shield-to-shield spacing, or read gap (RG), is crucial for improving resolution in the downtrack direction. However, further reduction is limited because the FL and PL require sufficient thickness to maintain TMR performance.

Another approach to improving downtrack resolution is to employ differential readers, which generate signals at transitions between recording bits. Representative examples include differential dual spin-valve (DDSV) readers [5] and synthetic antiferromagnetic (SAF)-FL readers [6]. An SAF-FL reader consists of two FLs that are antiferromagnetically coupled through a negative exchange interaction induced by a spacer material such as Ru and are biased in opposite directions by SAF side shields. DDSV and SAF-FL readers require symmetric parameters between the dual SVs and FLs, respectively, which is highly challenging due to their structural complexity. Another difficulty of differential readers is their intrinsically poorer performance in reading long patterns compared with conventional readers because of the zero response away from transitions.

The structure proposed in this study is an intentionally unbalanced differential reader (Fig. 1). It is based on the SAF-FL reader, but the magnetic volume of the second FL (FL2) is smaller than that of the first FL (FL1), forming a synthetic ferrimagnetic (SyF)-FL. Unlike conventional SAF-FL readers, which also require an SAF configuration for the side shields, the SyF-FL reader can employ the same side shields as a conventional single-FL reader, which enhances the asymmetry between FLs due to the bias field. As a result, the proposed structure becomes relatively simple, requiring only the addition of a spacer and FL2 to a conventional single-FL reader.

Figures 1(b) and (c) show schematics of the operation. At a bit transition, the magnetizations of FL1 and FL2 rotate further in the +*y* and -*y* directions, respectively, to maintain their antiparallel configuration. At the center of a recording bit, the magnetization rotation is relatively suppressed. However, a finite rotation still exists because it is energetically favorable for the system for FL1, with its larger magnetic volume, to align with the magnetic field direction. Consequently, the waveform of the SyF-FL reader becomes quasi-differential, with a conventional waveform that is decorated with additional signals at transitions. In this study, we confirmed this behavior through micromagnetic simulations.

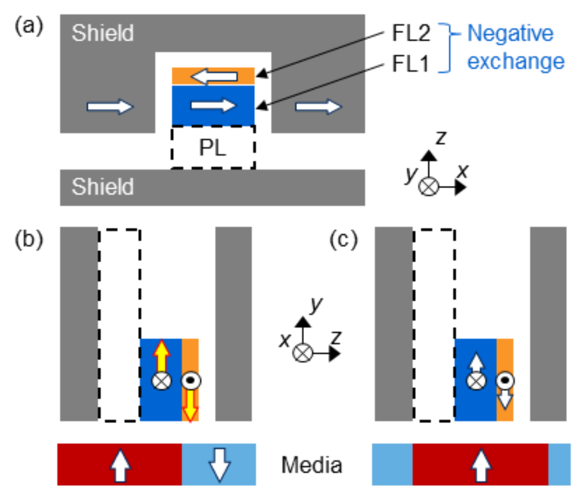

Fig. 1. Schematics of the structure and operation of a synthetic ferrimagnetic (SyF)-free layer (FL) reader. (a) Top view. Two FLs, FL1 and FL2, are antiferromagnetically coupled through a negative exchange interaction induced by the spacer. White arrows indicate the magnetization directions. The pinned layer (PL) is not included in this study. (b), (c) Cross-sectional views. Magnetization rotations are enhanced at transitions between recording bits (b) and are maintained at the center of recording bits because of the ferrimagnetic nature (c).

II. Simulation Parameters

To confirm the feasibility of the SyF-FL quasi-differential reader, we performed micromagnetic simulations using parameters similar to those in recent studies [4], [7]. The width and height of the FLs were 20 nm. The thicknesses of FL1 and FL2 were t_1 = 10 nm and t_2 = 4 nm, respectively, and were varied within the range of 10 nm < $t_1 + t_2$ < 14 nm. The saturation magnetization, $\mu_0 M_s$ of both FLs was 1.0 T. The SyF-FL was shielded as shown in Fig. 1, with gaps of 4 nm and 10 nm at the top and bottom, respectively. The bottom gap provides space for the PL; however, because the PL was not included in our model, the FL1 magnetization in the *y*-direction, m_y(FL1), was used as the signal.

The waveform was calculated by statically moving the recording media. The media model was assumed to have a simple rectangular shape with zero transition length, as shown in Fig. 1. Thermal magnetic noise was calculated separately without the media at 300 K. The root mean square values of the signal (S_{rms}) and noise (N_{rms}) were obtained from FFT analysis.

III. Simulation Results

The obtained waveforms for bit lengths of 60, 20, and 10 nm are shown in Fig. 2. A conventional single-FL reader (*i.e.*, (t_1, t_2) = (10, 0)) waveform is also shown for reference. As expected, m_y(FL1) exhibits a larger amplitude at transitions, while remaining comparable to the single-FL value at the center of the bit. The FL2 magnetization tends to align antiparallel to that of FL1 and exhibits a larger amplitude. These results clearly indicate that FL2 induces a quasi-differential waveform.

When considering shorter bit lengths (Fig. 2(b) and (c)), the advantage of the differential function becomes evident. At bit lengths of 20 and 10 nm, although both the SyF- and single-FL readers exhibit similar waveform shapes, the amplitude is clearly larger for the SyF-FL reader.

Next, we discuss the dependence of downtrack resolution and N_{rms} on the FL thickness ratio, where the downtrack resolution is defined as the ratio of S_{rms} between bit lengths of 20 and 60 nm.

As shown in Fig. 3(a), the resolution is well described by the thickness ratio. This behavior differs from that of conventional readers, in which the FL thickness and RG are dominant factors. At a ratio of 0.4, the resolution exhibits a step-like increase. Although not shown here, below this ratio the differential peak is not apparent. A ratio higher than 0.5 leads to hysteretic behavior during reading, which is not suitable for readers.

Increasing the thickness ratio also enhances N_{rms} (Fig. 3(b)). We observed a large fluctuation in the FL2 magnetization and attribute it to the origin of the enlarged N_{rms}.

IV. Conclusions

We confirmed the quasi-differential operation of the SyF-FL reader. Its simple structure and finite signal at the center of a bit provide a practical route to introducing differential functionality for higher recording density, particularly in the downtrack direction. The increase in thermal magnetic noise represents a trade-off with the improved resolution and should be suppressed through optimization of the reader stack or shield configuration.

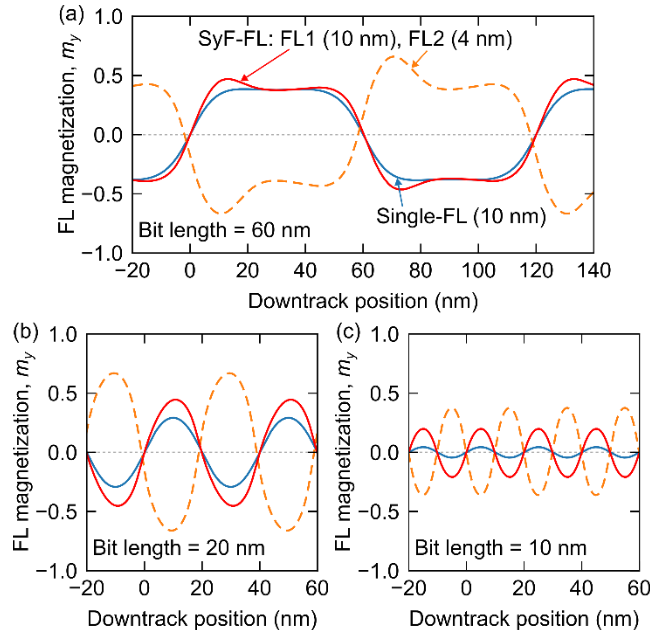

Fig. 2. Waveform of the synthetic ferrimagnetic (SyF)-free layer (FL) reader compared with that of a single-FL reader for various bit lengths. The SyF-FL reader exhibits differential signals at transitions.

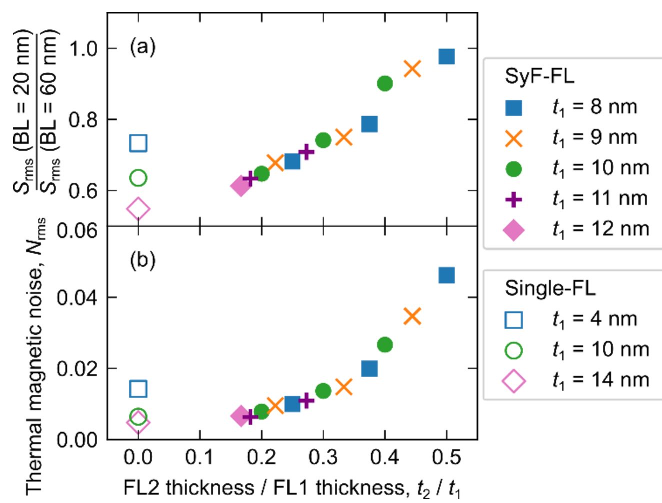

Fig. 3. Thickness ratio dependence of (a) resolution defined as the signal (S_{rms}) ratio between short and long pattern and (b) thermal magnetic noise (N_{rms}). Note that the total thickness and read gap vary between data points.

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